



Stage 4 Quality Loop Repair Stress Update

Uncertainty calibration across clustered data and segmentation stress cases

Research Presentation Program

Department of Statistics & Data Science

2026-08-27



香港中文大學
The Chinese University of Hong Kong



Clustered outcomes separate center and individual variation

$$Y_{ij} = \beta_0 + \beta_1 T_{ij} + u_j + \varepsilon_{ij}, \quad u_j \sim N(0, \tau^2), \quad \varepsilon_{ij} \sim N(0, \sigma^2)$$

Center effects set $\rho = \tau^2 / (\tau^2 + \sigma^2)$, so interval calibration depends on cluster count.

Model components

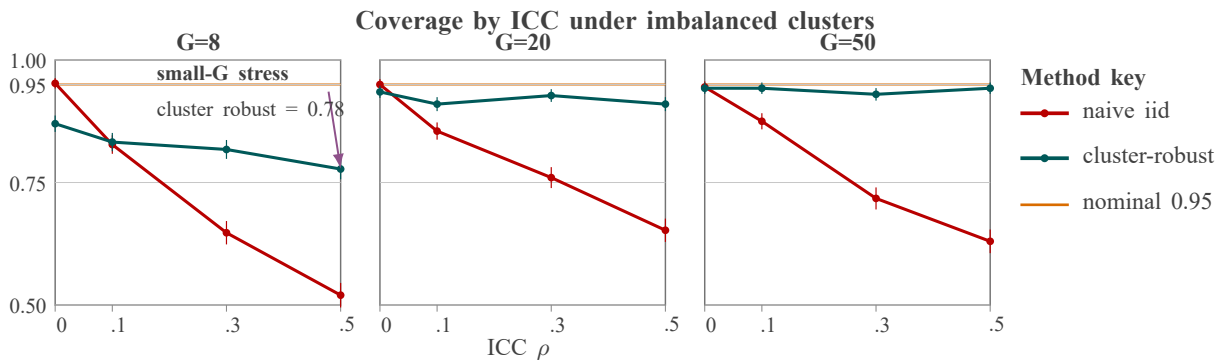
- clustered outcome equation
- center random effect
- individual error

Interpretation

The source model makes center-level dependence explicit before interval comparison.



Cluster-robust intervals recover coverage as ICC increases



Data source: 400 simulation replicates per cell; y-axis shows 95% interval coverage.

The source simulation shows method-specific coverage under imbalanced clusters.

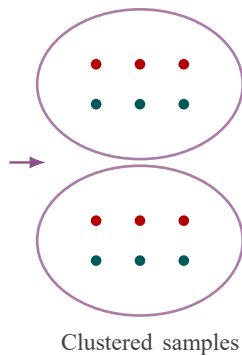
Synthetic clustered-data example; intervals compared on coverage, width, and bias.

Simulation varies clustering before interval procedures are compared

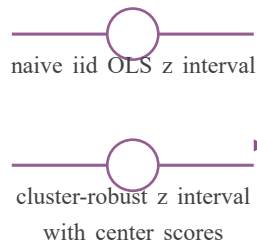
DGP knobs

- centers $G = 8, 20, 50$
- ICC $\rho = 0, .1, .3, .5$
- balanced or imbalanced clusters

Clustered samples



Interval procedures

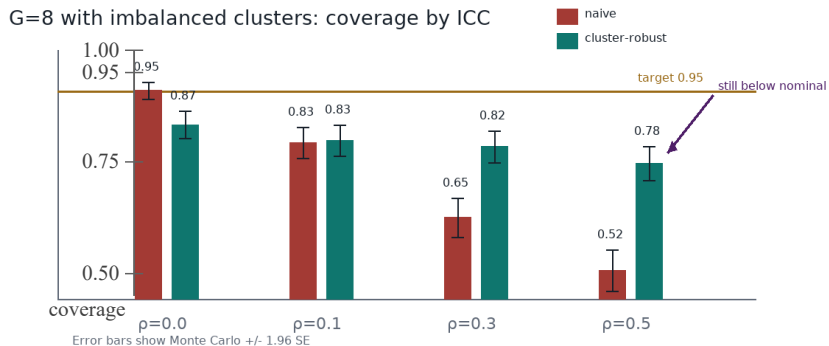


Coverage diagnostics

- Coverage target 0.95
- mean interval width
- treatment-effect bias

The connector direction encodes data generation before interval estimation and coverage diagnostics.

Small-G settings remain anti-conservative after robustification



The failure region is adjacent to the diagnostic explanation instead of hidden in a note.

Negative evidence is completed; CR2 and wild cluster bootstrap remain the next test.

Next experiment tests whether batch selection reduces fragile coverage

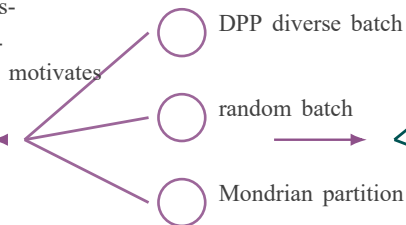
Failure evidence

G=8, ICC=.5 and imbalanced clusters remain below nominal coverage.

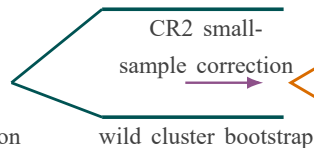


Small-G limit

Manipulate sampling



Comparator arms



Decision rule



Go if coverage $\geq .94$ with controlled width; no-go stratifies by Mondrian cell.

Coverage evidence determines which batch-query strategy to validate next; CR2 and wild bootstrap remain proposed comparators.



Same-case panels keep the segmentation error interpretable

Research direction

Segmentation robustness: independent visual failure analysis.

